

VMware to KubeVirt

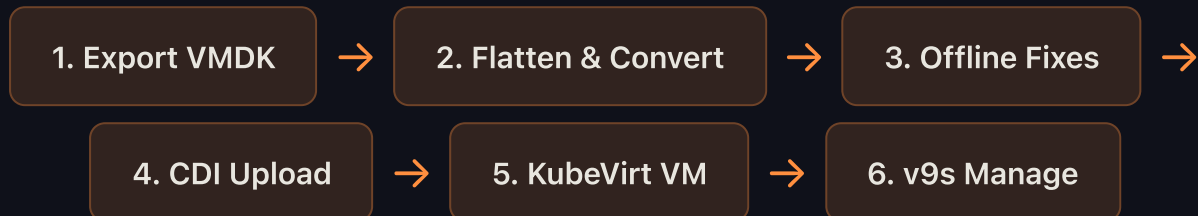
The Complete Migration Journey

From VMware VMDK to a fully managed KubeVirt VM — migration with h2kvm, management with v9s. End-to-end automation with first-boot success.

h2kvm + v9s — Migrate & Manage

The Journey

Six stages from VMware to production KubeVirt.



1

Export from VMware

Export VMDK via govc, OVF Tool, or vSphere HTTP API. h2kvm supports all export methods with automatic format detection.

2

Flatten & Convert

Flatten sparse/split VMDKs into a single file. Convert to qcow2 with optional compression. Checksum validation for integrity.

3

Automated Offline Fixes

GuestKit engine mounts the image and applies deterministic fixes: GRUB repair, VirtIO drivers, fstab stabilization, initramfs rebuild, network config, serial console.

4

CDI Upload to Kubernetes

Automatic CDI (Containerized Data Importer) upload creates a PersistentVolumeClaim with the disk image. qcow2→raw conversion handled by CDI.

5

KubeVirt VM Creation

VirtualMachine manifest generated with CPU, memory, disk, and network configuration. VM boots successfully on first attempt.

6

Ongoing Management with v9s

Web-based management: console access, metrics monitoring, snapshots, storage, network, and host visibility. Full day-2 operations.

Step 1–5: Migration

One command to migrate a VMware VM to KubeVirt.

Single Command Migration

```
$ h2kvmctl \
  --cmd local \
  --vmdk /vmware/ubuntu-server2504.vmdk \
  --output-dir /output \
  --flatten \
  --to-output ubuntu-server2504.qcow2 \
  --out-format qcow2 \
  --fstab-mode stabilize-all \
  --regen-initramfs \
  --serial-console \
  --compress
```

What Happens Automatically

Stage	Action	Result
Flatten	Merge sparse VMDK extents	Single monolithic disk file
Convert	VMDK → qcow2	KVM-native format with compression
Inspect	GuestKit mounts & detects OS	Ubuntu 25.04, GRUB2, ext4, no LVM
Fix GRUB	Reinstall bootloader for VirtIO	Boot from virtio-blk instead of SCSI
Fix initramfs	Add VirtIO modules	Kernel loads virtio_blk, virtio_net
Fix fstab	Device names → UUID	Stable mounts regardless of device order

Serial
console

Enable ttyS0

Headless management via serial

Validate

QEMU smoke test

VM boots successfully

KubeVirt Deployment

```
$ demo-kubevirt.sh ubuntu-server2504.vmdk  
# Pipeline: VMDK → flatten → offline fixes → CDI upload → VM
```

Step 6: Management

VM running on KubeVirt, managed through v9s web UI.

Dashboard View

After migration, the VM appears in the v9s dashboard with real-time metrics. The dashboard shows KubeVirt cluster status, host resources, and recent events.



Cluster Overview

1 VM running on KubeVirt v1.8.0 / K3s v1.34.5. Host CPU at 66%, memory at 52%, disk at 78%. Uptime 2d 21h. Real-time metrics refresh.



VM Status

ubuntu-server2504 showing as "Running" with green status badge. Namespace: default. Ready: True. Full lifecycle controls available.

VM Details Panel

Click the VM to open the 10-tab detail panel with complete management capabilities:



Summary

Status, namespace, ready state, node placement, resource allocation, action buttons.



Console

VNC graphical + serial + SSH. Ctrl+Alt+Delete sendkey. Fullscreen mode.



Metrics

Real-time CPU, memory, network I/O, disk I/O via WebSocket streaming.

Host Visibility

Complete hardware and system inventory of the KubeVirt host.

System Information

Hostname	NLDW3-4-31-12
OS	AlmaLinux 9.7 (Moss Jungle Cat)
Kernel	5.14.0-611.41.1.el9_7.x86_64
CPU	Intel Xeon E3-1230 v6 @ 3.50GHz
Cores	8 / 8 threads
Vendor	FUJITSU PRIMERGY RX1330 M3
KubeVirt	v1.8.0 on K3s v1.34.5

Hardware Inventory

RAM	16 GB DDR4 (Kingston, 2400 MHz)
PCI Devices	15+ devices (Intel Host Bridge, USB, SATA, NIC)
Disk	/dev/sda4 (219 GB ext4, 78% used)
Boot	/dev/sda2 (1 GB xfs, 34% used)
EFI	/dev/sda1 (600 MB vfat, 2% used)
Services	55+ systemd services tracked

Operational Capabilities



Snapshots



Storage



Network

Create point-in-time backups.
Restore on demand. API version auto-detection.

Manage disks, mount ISOs, view storage classes.
Duplicate protection.

Add interfaces (pod/multus).
Bridge, masquerade, SR-IOV bindings.

Migration Results

Real-world migration of Ubuntu Server 25.04 from VMDK to KubeVirt.

1

Command to Migrate

100%

First-Boot Success

0

Manual Fixes Required

Running

VM Status on KubeVirt

Metric	Value
Source Format	VMware VMDK (Ubuntu Server 25.04)
Target Platform	KubeVirt v1.8.0 on K3s (AlmaLinux 9.7)
Automated Fixes	GRUB2, initramfs, fstab (UUID), serial console
Boot Test	✓ QEMU smoke test passed
CDI Upload	✓ PVC created, image uploaded
VM Status	✓ Running, Ready: True
Web Management	✓ v9s: console, metrics, snapshots, host info

From VMware to KubeVirt in Minutes

h2kvm handles the migration. v9s handles the management.
Zero downtime. First-boot success. Complete visibility.

